

GANESH SHARMA

AI Engineer & Python Backend Developer
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GitHub | LinkedIn | Portfolio Website

SUMMARY

Python Backend & AI Engineer with experience building scalable backend architectures and AI-integrated applications. Proficient in developing high-performance RESTful APIs using FastAPI, PostgreSQL, and Redis, alongside production-ready RAG pipelines utilizing LangChain and vector databases. Passionate about combining robust software engineering with modern AI solutions.

TECHNICAL SKILLS

Languages: Python, C++

Back-End Engineering: FastAPI, PostgreSQL, SQLAlchemy, Redis, RESTful API Design

AI & Retrieval Engineering: LLM APIs, LangChain, LangGraph, ChromaDB, RAG Pipelines

Data/ML: Pandas, NumPy, Matplotlib, Scikit-Learn, ML Fundamentals

Dev Tools: Git, GitHub, Docker, VS Code, Agentic IDEs

EDUCATION

Bharat College of Engineering

B.E. in Computer Science & Engineering (AI & ML) | University of Mumbai

Mumbai, India

2022–2026

PROJECTS

Full-Stack Blog Platform | *FastAPI, PostgreSQL, SQLAlchemy, Pydantic, Jinja2, CRUD*

- Architected a modular full-stack blog platform using FastAPI, PostgreSQL, and SQLAlchemy, isolating API routing, business logic, database models, and validation schemas to maximize system maintainability.
- Designed and implemented RESTful APIs with CRUD capabilities, executing robust data validation via Pydantic and robust token management via JWT-based authentication and role-based authorization.
- Engineered relational database schemas in PostgreSQL, leveraging SQLAlchemy ORM to manage complex entity relationships and optimize data persistence storage for efficient runtime queries.

AI Document Summarizer | *FastAPI, LangChain, LangGraph, LLM APIs, PyPDFLoader, TextLoader, WebBasedLoader*

- Engineered an AI-powered document summarization service using FastAPI, providing scalable RESTful endpoints for multi-format uploads, including PDF files, text documents, and web pages.
- Architected a modular processing pipeline for text extraction, prompt engineering, and LLM inference, utilizing LangChain to maintain a unified, easily extensible model interface.
- Implemented robust backend infrastructure, combining Pydantic request validation, comprehensive error handling, and structured API responses to ensure system reliability and seamless data delivery.

MITRA AI Chatbot | *OpenAI/Gemini/Groq/Mistral APIs, LangChain, FastAPI, Ollama*

- Engineered MITRA, an AI assistant via FastAPI, architecting a unified backend that integrated proprietary and locally hosted LLMs (OpenAI, Gemini, Groq, Mistral, Ollama) for flexible model selection.
- Developed modular backend services for prompt orchestration, model routing, conversation history management, and response generation to optimize code maintainability and system extensibility.
- Investigated 8-bit quantized language model deployment on consumer-grade GPUs, optimizing local LLM inference efficiency and publishing the findings in a peer-reviewed research paper.

COURSES & CERTIFICATIONS

- Harvard CS50's Introduction to Programming with Python (edX)
- Machine Learning Bootcamp Certification — IIT Bombay
- Machine Learning Crash Course — Google Developers
- Google's AI Essential Course — Coursera

RESEARCH & PUBLICATIONS

MITRA AI Chatbot Assistant: Using 8-Bit Quantized Large Language Models on Consumer-Grade GPUs

Published in *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*

Volume 14, Issue III, pp. 5059–5065, ISSN: 2321-9653.